4-12-18(E)

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

Nov – Dec 2018

Max. Marks: 100	**Duration: 3 hrs**
Class: SYBTECH	Semester: IV
Name of the Course: **Microprocessor**	Branch : Computer
Course Code: **UCEC402**	

Instructions:

(1) All Questions are Compulsory

(2) Draw neat diagrams

(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1	Design an 8086 based system with following specification: a) 8086 working at 8 MHz. b) 64 KB EPROM using 32 KB devices. c) 32 KB SRAM using 16 KB devices. d) 1 input, 1output port(both 16 bit) interrupt devices.	20M
Q2 (a)	Write a program in 8086 to transfer block of data using string instructions. **OR** Write a program in 8086 to find whether string is palindrome or not.	10 M
Q2(b)	Write short note on Mixed language programming and its objectives.	10 M
Q3 (a)	Explain segment translation mechanism in detail for 80386 processor.	10 M
Q3(b)	Compare real mode and protected mode of 80386 processor.	10 M
Q4 (a)	Draw block diagram of Pentium Processor and explain its superscalar operation. **OR** How branch prediction is done in Pentium.	10 M
Q4(b)	Compare Intel i3,i5 processor	10 M

Q5	Write short notes on (**Any 2**) a)8237 DMA b) Math Coprocessor c)8259 interrupt controller d)8255	20 M

8.12.2018 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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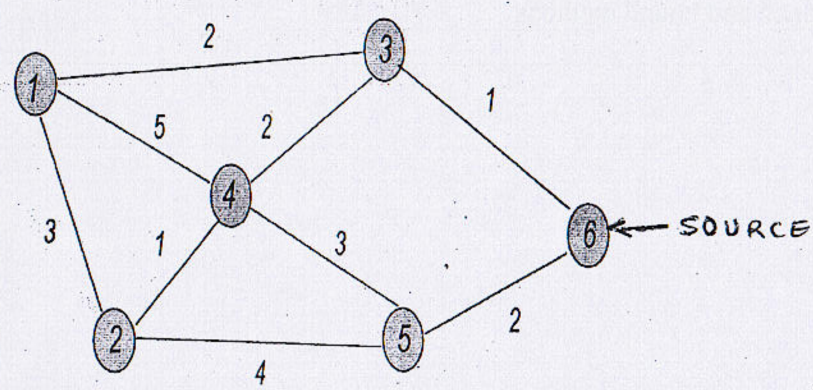
End Semester Exam
Nov – Dec 2018

Max. Marks: 100 | **Duration: 03 hrs**
Class: S Y BTech | Semester: IV
Name of the Course: Analysis of Algorithm | Branch: Computer
Course Code: UCEC403

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	(i) Define O , Omega and Theta notations of algorithm complexity. (ii) Explain the terms Time Complexity and Space Complexity	05 05
Q 1 (b)	Apply Finding minimum and maximum algorithm by divide and conquer strategy for following numbers. Show output of each pass with explanation. 45 , 56, 12, 89, 79, 34, 93, 25, 67	10
Q2 (a)	Discuss quicksort algorithm. Write pseudocode for the same. Explain time complexity of this algorithm.	10
Q2 (b)	Differentiate between Greedy methods, dynamic programming methods and backtracking methods	10
Q3 (a)	 Solve for above graph for shortest path using Bellman Ford algorithm from vertex 6 Source. Show all steps. OR Solve for above graph for shortest path using Dijkstra's algorithm from	10 10

	vertex 6 Source. Show all steps.	
Q3 (b)	Write algorithm for Minimum spanning tree by Prim's method. Explain with small example.	10
Q4 (a)	Solve following sum of subsets problem using backtracking method. M=30, W ={5, 10, 12. 13, 15, 18} . Show all steps. OR Explain n-queens problem and write the algorithm for the same with an example	10 10
Q4 (b)	Solve following knapsack instance by dynamic programming. Capacity =8 and N = 4. Let Pi & Wi are as shown in table. <table><tr><th>i</th><th>Pi</th><th>Wi</th></tr><tr><td>1</td><td>1</td><td>2</td></tr><tr><td>2</td><td>2</td><td>3</td></tr><tr><td>3</td><td>5</td><td>4</td></tr><tr><td>4</td><td>6</td><td>5</td></tr></table> OR Explain at least TWO String matching techniques with algorithm.	10 10
Q5	Write short notes on any TWO 1) All pair Shortest path algorithm 2) Job sequencing with deadlines 3) Recursion tree method of algorithm analysis 4) Travelling salesman problem 5) Branch and bound methods	20

13·12·2018 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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Nov Dec 2018

Max. Marks:100 | **Duration:3 hrs**
Class: SY | Semester: IV
Name of the Course: Digital Communication Network | Branch: COMPUTER
Course Code:UCEC405

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		**Max. Marks**
Q 1 (a)	Explain the importance of communication. Draw the diagram to explain the elements of digital communication systems	10
Q 1 (b)	What are the types of electronics communication? Explain 8 point difference between Analog and Digital communication	10
Q2 (a)	Explain Huffman Coding problem with an appropriate example	10
Q2 (b)	What is meant by multiplexing? Explain CDM in detail OR Differentiate between TDM ,FDM and WDM	10 10
Q3 (a)	Write a short note on PCM and ADM OR Write short notes on PSK and FSK	10 10
Q3 (b)	What is meant by the terms Information, Information Rate, and Uncertainty? Discuss about the term Entropy with a suitable example	10
Q4 (a)	List and explain OSI layers in detail OR Explain all types of topology in brief	10 10
Q4 (b)	What do you mean by LAN, WAN and MAN. Explain with suitable real time examples	10
Q5 (a)	Write a short note on Go back N flow control mechanisms	10
Q5 (b)	Explain ALOHA Protocol in detail What is meant by Pure ALOHA OR What do you understand by Persistence methods. Explain 1 persistent, Non Persistent and P persistent methods	10 10